ISSEY SONE

Citizenship: United States
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EDUCATION

University of Chicago — M.S. in Statistics *Sep 2026 - Present*

University of Toronto — B.Sc. in Statistics *Sept 2022 - June 2026*
Graduated with High Distinction
Probability and Stochastic Processes, Machine Learning, Statistical Inference, Time Series, Regression Analysis, Topology, Real Analysis, Methods of Multivariate Statistics, Algorithms and Data Structures

RESEARCH EXPERIENCE

Large Deviations and Gauss Hypergeometric Function Sep 2025 - Present
Supervisors: Prof. Vladimir Vinogradov, and Prof. James Bremer Jr. UofT

- Proving, and numerically verifying, new formulas for sums of zero-modified geometric distributions using Hypergeometric function ${}_2F_1$.
- Developing numerical methods to compute large deviation rate functions for sums of zero-modified geometric random variables and their extended families.
- Deriving closed-form rate functions where possible, and comparing their accuracy of approximation.
- Used a new formula to derive novel approximations of the Hypergeometric function ${}_2F_1$, including an explicit derivation of the special case where the final parameter is fixed to 1.

Evaluation of Impact of a Digital Multilingual STEM Resource Jan 2025 - May 2025
Supervisors: Prof. Sohee Kang and Prof. Emmanuelle Le Pichon-Vorstman UofT

- Designed and conducted quantitative analyses on multilingual STEM learning data, applying survival analysis, statistical hypothesis testing, and Markov models to evaluate student language usage behavior.
- Created 15+ tailored visualizations (Kaplan–Meier plots, transition diagrams, comparative charts) to support interpretation of results and dissemination in academic contexts.
- Explored multiple methodological approaches (e.g., survival curves, state-transition modeling, regression) to identify robust predictors of student retention across 1K+ records.

PRESENTATIONS/TALKS

Introductory Large Deviations August 2025
Fields Institute, University of Toronto
Large Deviation Theory, Information Theory

TEACHING

An Introduction to Probability (STAB52) Summer 2026
Teaching Assistant at UofT (Instructor: Dr. Sotirios Damouras)
Introduction to Statistics for the Social Sciences (STAB23) Summer 2026
Teaching Assistant at UofT (Instructor: Dr. Vladimir Vinogradov)
An Introduction to Statistics (STAB57) Winter 2026
Teaching Assistant at UofT (Instructor: Dr. Shahriar Shams)
Statistics I (STAB22) Winter 2025, Fall 2025, Winter 2026
Teaching Assistant at UofT (Instructor: Dr. Sohee Kang, Dr. Mahinda Samarakoon)
Methods of Multivariate Statistics (STAD37) Fall 2025
Teaching Assistant at UofT (Instructor: Dr. Shahriar Shams)

COMMUNITY INVOLVEMENT

UTSC SDG Data Challenge May 2025
Lead Volunteer, Mentor

- Coordinated event logistics and volunteer activities to help ensure the smooth running of the UTSC SDG Data Challenge.
- Mentored student participants by answering technical and organizational questions, and providing guidance on project approaches.

Association of Computer and Mathematical Sciences (AMACSS) Sep 2024 – Present
Course Representative, Director of Statistics

- Director of Statistics (2023 – 2024): Organized, coordinated, and oversaw academic initiatives to run review seminars for statistics courses, supervising their progress, and managing multiple teams.
- Course Representative (Sep 2024 – May 2025): Prepared material and ran midterm and final review seminars for STAB57 – Introduction to Statistics, and STAC67 – Regression Analysis.

UTSC Rock Climbing Club June 2025 – Present
Vice President

- Obtained CWA certification to allow the club to operate, enabling regular climbing sessions for members.
- Coordinated logistics and ensured compliance with university guidelines for recreational activities.

WORK EXPERIENCE

University of Toronto Jun 2025 - Aug 2025
Rshiny Analyst, Supervisor: Prof. Shahriar Shams

- Developed nine interactive RShiny dashboards to visualize key concepts in multivariate statistics (e.g., PCA, clustering, Confidence Intervals) for STAD37 – Multivariate Analysis.
- Enhanced learning accessibility by integrating dynamic controls and intuitive UI features, enabling students to explore statistical concepts and visuals.
- Deployed dashboards on university servers in collaboration with the course instructor, ensuring alignment with curriculum and broad student access.

BlueArc AI May 2024 - Sep 2024
Data Science Intern

- Developed and automated scalable data pipelines (APIs, scraping, data warehouses) and applied NLP methods to extract key insights from large unstructured datasets.
- Improved fraud-detection scoring accuracy by 55% and increased profile completeness by 25% through data enrichment and feature engineering.

SKILLS/HOBBIES

Programming Languages	Python, R, Java, C, C#, SQL, LaTeX
Hobbies	Rock climbing, Badminton, Cooking